

DexSynRefine: Synthesizing and Refining Human-Object Interaction Motion for Physically Feasible Dexterous Robot Actions

Hyesung Lee^{1,2,*}, Hyunwoo Jung³, Si-Hwan Heo¹, Sungwook Yang¹

¹Korea Institute of Science and Technology, ²KAIST, ³Hanyang University

Abstract: Learning dexterous manipulation from human-object interaction (HOI) data is a scalable alternative to teleoperation, but HOI demonstrations are sparse and provide only kinematic motion that is not directly executable under embodiment mismatch and contact-rich dynamics. We present **DexSynRefine**, a framework with three coupled components: HOI-MMFP, a task and object initial state-conditioned motion manifold primitives that synthesizes coordinated hand-object trajectories from sparse HOI demonstrations; a task-space residual RL policy that physically grounds the synthesized reference while inheriting its kinematic structure; and a contact-and-dynamics adaptation module that enables sim-to-real transfer from proprioceptive history. Across five dexterous manipulation tasks spanning pick-and-place, tool use, and object reorientation, our task-space residual policy outperforms prior action-representation baselines in simulations and transfers to a real robot on all five tasks, improving over kinematic retargeting by 50–70 percentage points. Project page: <https://dexsynrefine.github.io/>

Keywords: Dexterous Manipulation, Motion Synthesizing, Learning from Human Data, Reinforcement Learning

1 Introduction

Recent advances in imitation learning [1–4] have significantly improved multi-fingered dexterous manipulation, enabling increasingly dexterous and contact-rich robot behaviors learned from large-scale teleoperation datasets [5–8]. However, collecting such datasets remains expensive, labor-intensive, and difficult to scale, especially for dexterous tasks that require coordinated finger motion, accurate contact timing, and stable object control. Because human hands naturally produce exactly these contact-rich behaviors during everyday object interaction, a growing line of work has turned to more accessible recordings of human-object interaction (HOI)—ranging from Internet videos to motion-capture sessions [9–12]—as a substitute source of supervision.

Yet converting human HOI data into executable dexterous robot behavior remains fundamentally difficult. Human demonstrations primarily provide geometric or kinematic motion, while

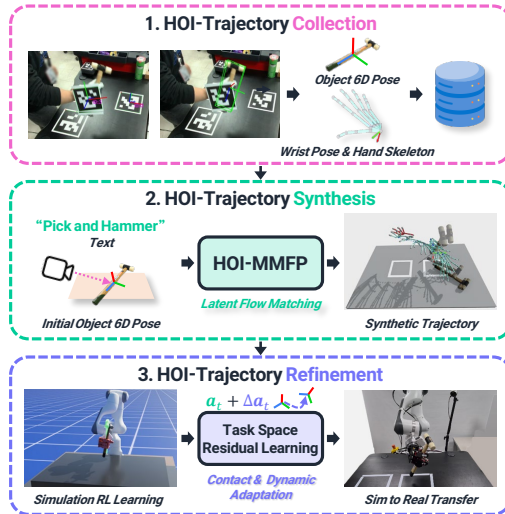


Figure 1: **DexSynRefine** synthesizes kinematic trajectories from a few HOI demonstrations conditioned on the initial object pose and task, then refines them into physically feasible dexterous actions via residual RL.

successful manipulation depends on contact forces,

friction, and other aspects of contact-rich dynamics that are not directly observed. In addition, human and robot hands differ substantially in kinematic structure, actuation, and contact geometry, creating a large embodiment gap between observed human motion and feasible robot action. Recent work has mitigated this issue by using human or object motion as references for goal-conditioned reinforcement learning [13–18]. However, these methods still rely on suitable reference interactions for each task instance. Since HOI datasets are inherently sparse, such references may not exist when the task configuration or initial object state departs from the training distribution. The key challenge is therefore not only to physically refine a given human prior, but also to construct task and state-generalizable interaction priors from sparse HOI data.

We address this challenge with **DexSynRefine**, a framework that learns task and object initial state conditioned HOI motion priors from sparse demonstrations and grounds them into deployable dexterous actions. We instantiate the prior as HOI-MMFP, an extension of Motion Manifold Flow Primitives (MMFP) [19, 20] that synthesizes interaction trajectories conditioned on task identity and initial object state. The synthesized trajectories are then refined through task-space residual reinforcement learning to produce physically feasible behavior in simulation under embodiment mismatch and contact-rich dynamics. While the residual policy yields physically feasible behavior in simulation, real-robot deployment introduces an additional challenge object properties and contact states observable in simulation are unavailable from onboard sensing. Prior adaptation methods such as RMA [21, 22] infer hidden dynamics from proprioceptive history, but dexterous execution also depends on accurately responding to rapid finger-object contact transitions. DexSynRefine therefore couples a contact estimator with a dynamics latent inferred from proprioceptive history, improving real-world robustness.

Our contributions are threefold: (1) a task and state-conditioned generative prior for coupled hand-object interaction trajectories, (2) a residual RL formulation that physically grounds synthesized HOI motion under embodiment and contact constraints, supported by a systematic comparison of action representations identifying task-space residuals as the most effective formulation, and (3) a contact and dynamics aware adaptation mechanism for robust sim-to-real transfer of dexterous manipulation policies.

2 Related Works

Augmenting HOI Motion Prior with Goal Conditioned RL. A promising direction to bridge kinematic human motion and dynamic robot execution is to use HOI demonstrations as priors and learn an absolute or residual policy via reinforcement learning. Existing formulations vary in their action representation: Zhao et al. [15] learn a joint-space residual added to a retargeted primitive; Li et al. [14] pre-train an imitation policy and then fine-tune a residual policy for physical interaction; Chen et al. [13] adopt a hierarchical design with a high-level wrist-imitation policy and a low-level RL policy that mixes task-space wrist residuals with absolute finger commands; and Lum et al. [18] learn an absolute joint-action policy on top of a geometric fabric controller, using human hand kinematics only in the pre-manipulation phase.

However, these methods rely on sparse HOI demonstrations, and which action representation is most effective for HOI-conditioned RL has not been systematically studied. They also require a matching reference at deployment, which breaks under out-of-distribution initial object poses. Concurrent HOI generative models provide motion priors but focus on unconditional or text-conditioned generation, lacking the joint conditioning on task and initial object pose required for online deployment. We therefore propose Human-Object Interaction Motion Manifold Flow Primitives (HOI-MMFP), which jointly synthesizes coherent hand and object trajectories from initial conditions for dense RL guidance. Our systematic comparison further identifies task-space residual RL on both wrist and fingers as the most effective formulation for bridging the embodiment gap.

3 Method

3.1 Task Setup

We consider the problem of transferring sparse human-object interaction (HOI) demonstrations into executable robot behavior for dexterous manipulation. Each demonstration is represented as

$$\tau = \{(\mathbf{w}_t, \mathbf{h}_t, \mathbf{o}_t)\}_{t=1}^T, \quad (1)$$

where $\mathbf{w}_t \in \text{SE}(3)$ and $\mathbf{o}_t \in \text{SE}(3)$ denote the wrist pose and object pose in the world frame, respectively, and $\mathbf{h}_t \in \mathbb{R}^{K \times 3}$ denotes the K hand keypoints expressed in the wrist frame. Each trajectory is associated with a task condition $\mathbf{c} = (\mathbf{u}, \mathbf{o}_0)$, where $\mathbf{u}$ is a task embedding and $\mathbf{o}_0$ is the initial object pose.

Given a sparse HOI demonstration set $\mathcal{D} = \{(\tau^{(i)}, \mathbf{c}^{(i)})\}_{i=1}^N$, our goal is to learn a dexterous robot policy that generalizes to unseen initial object poses. DexSynRefine addresses this problem in three stages: (1) object-centric augmentation of sparse HOI trajectories, (2) task- and state-conditioned HOI motion prior synthesis via HOI-MMFP, and (3) residual policy learning with deployment-oriented contact and dynamics adaptation.

3.2 HOI Motion Collection and Object-Centric Augmentation

We collect seven HOI demonstrations per task for five dexterous manipulation tasks (*Reorient Pringles Bottle*, *Pick Up and Hammer*, *Pick and Pour Watering Can*, *Pick and Place Book*, *Pick and Place Bowl*) spanning pick-and-place, tool use, and object reorientation, using a multi-camera setup with FoundationPose [23] and Manus Gloves, varying only the object’s initial yaw.

To generalize from such sparse data, we apply object-centric augmentation: initial object poses are perturbed by planar $\text{SE}(2)$ transformations, with the reaching phase interpolated and the manipulation phase rigidly transformed to preserve hand-object structure. For symmetric objects, we additionally randomize rotations around the symmetry axis. This yields approximately 300 augmented trajectories per task, each resampled to a fixed horizon of $T = 220$. Details are provided in Appendix A.

3.3 HOI-MMFP: Task-Conditioned HOI Motion Prior Synthesis

HOI-MMFP extends Motion Manifold Flow Primitives (MMFP) to human-object interaction by synthesizing trajectories conditioned on task identity and initial object state. The overall training pipeline of HOI-MMFP is illustrated in Figure 2(a).

The training of HOI-MMFP proceeds in two decoupled stages: manifold learning and conditional latent flow matching. In the first stage, given an augmented trajectory τ^{aug} , we learn a latent HOI motion manifold using a Transformer-based autoencoder, where the encoder maps τ^{aug} to a latent code $\mathbf{z}^* = E(\tau^{\text{aug}})$ and the decoder reconstructs it as $\tilde{\tau} = D(\mathbf{z}^*)$. To ensure a well-structured and temporally continuous latent space, the autoencoder is optimized with a combined loss:

$$\mathcal{L}_{\text{AE}} = \mathcal{L}_{\text{recon}} + \eta \mathcal{L}_{\text{norm}} + \delta \mathcal{L}_{\text{smooth}}, \quad (2)$$

where $\mathcal{L}_{\text{recon}} = \|\tau^{\text{aug}} - \tilde{\tau}\|_2^2$ is the standard reconstruction error, and $\mathcal{L}_{\text{norm}} = \|\mathbf{z}^*\|_2^2$ regularizes the magnitude of the latent codes. Furthermore, to enforce kinematic continuity across the manifold, we introduce a smoothness regularization $\mathcal{L}_{\text{smooth}}$. Specifically, we decode linearly interpolated latent codes $\tau_{\text{mix}} = D(\alpha \mathbf{z}^* + (1 - \alpha) \mathbf{z}_j)$, where $\mathbf{z}_j$ is a randomly permuted latent code within the batch and $\alpha \sim \mathcal{U}(\alpha_{\min}, \alpha_{\max})$. $\mathcal{L}_{\text{smooth}}$ then minimizes the squared temporal differences of these interpolated trajectories, ensuring smooth transitions between states.

To enable task-conditioned generation on the learned manifold, we train a conditional flow-matching model in latent space, using the task condition $\mathbf{c} = (\mathbf{u}, \mathbf{o}_0)$ defined in Sec 3.1, where $\mathbf{u}$ is extracted by a pretrained text encoder. Let $\mathbf{z}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ be Gaussian noise and let $\mathbf{z}_s = (1 - s)\mathbf{z}_0 + s\mathbf{z}^*$ for $s \sim \mathcal{U}(0, 1)$. A Diffusion Transformer [24] based vector field v_θ model is trained with

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{\tau^{\text{aug}}, \mathbf{z}_0, s} \left[\|v_\theta(\mathbf{z}_s, s, \mathbf{c}) - (\mathbf{z}^* - \mathbf{z}_0)\|_2^2 \right]. \quad (3)$$

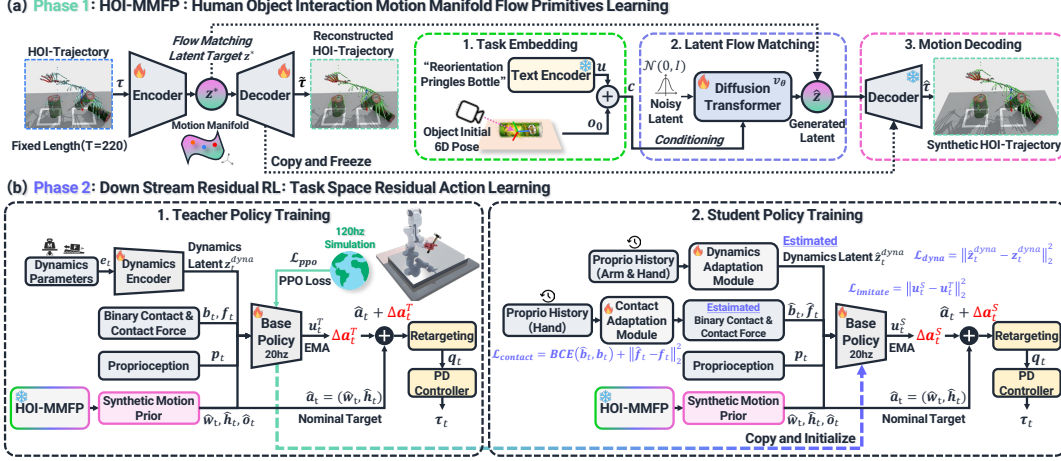


Figure 2: Overview of our framework. **(a) HOI-MMFP**: an autoencoder learns a motion manifold and a conditional flow matching model generates task-conditioned latents decoded into synthesized HOI trajectories. **(b) TaskSpace Residual RL**: a privileged teacher policy is trained in simulation, then a deployable student policy is distilled from proprioceptive histories with explicit contact estimation and latent dynamics adaptation.

At inference time, we integrate $dz_s = v_\theta(z_s, s, c) ds$ from $s = 0$ to $s = 1$ to obtain $\hat{z}$, and decode the synthesized HOI reference as $\hat{\tau} = D(\hat{z})$. The resulting structured trajectory $\hat{\tau} = \{(\hat{w}_t, \hat{h}_t, \hat{o}_t)\}_{t=1}^T$ serves as a task-consistent kinematic prior for downstream policy learning.

3.4 Task-Space Residual Action Learning for Physically Feasible Execution

Although HOI-MMFP produces kinematically plausible interaction trajectories, the synthesized motion is not directly executable on the robot due to embodiment mismatch and contact-rich physical constraints. We therefore learn a residual policy that refines the synthesized task-space references into physically feasible behavior, as illustrated in Figure 2(b). At the beginning of each episode, we randomize the initial object pose and query HOI-MMFP once to produce $\hat{\tau}$; subsequent control steps reuse this fixed reference.

Task Space Residual Action Integration. At each policy step t , the refinement policy predicts a task-space residual signal u_t , which is smoothed by an exponential moving average (EMA) filter:

$$\tilde{u}_t = \alpha u_t + (1 - \alpha) \tilde{u}_{t-1}, \quad (4)$$

where α is the smoothing coefficient. The residual action Δa_t (denoted Δa_t^T for the teacher and Δa_t^S for the student) is then obtained by integrating the filtered command and clipping it to predefined limits:

$$\Delta a_t = \text{clip}(\Delta a_{t-1} + s \cdot \tilde{u}_t \cdot \Delta t, \Delta a_{\min}, \Delta a_{\max}), \quad (5)$$

with s a scaling factor and $\Delta t = 0.05$ s. The clamping prevents the policy from diverging excessively from the synthesized HOI guidance. The refined task-space target is obtained by adding Δa_t to the wrist and fingertip components of the synthesized reference $\hat{\tau}_t$.

To bridge the frequency gap between the 20 Hz policy and the 120 Hz simulation, we apply quintic interpolation in task space and solve inverse kinematics—damped least-squares for the 7-DoF arm and an analytical solver for the 16-DoF hand—to obtain executable joint commands q_t at 120 Hz.

Privileged Teacher Policy Learning. We train a privileged teacher policy in simulation using PPO [25] with an asymmetric actor-critic formulation. The critic takes as input the full privileged state tuple $(s_t^{\text{priv}}, p_t^{\text{hand}}, p_t^{\text{arm}}, b_t, f_t, \hat{\tau}_t, e_t)$, which encompasses simulator-exclusive variables (e.g., joint velocities, torques, and Cartesian velocities), hand and arm proprioception, binary contact states, contact forces, the synthesized HOI reference trajectory, and intrinsic object dynamics parameters. Detailed state dimensions are provided in Appendix C.

The actor receives a compact dynamics representation $\mathbf{z}_t^{\text{dyna}} = \mu(\mathbf{e}_t)$ from a dynamics encoder $\mu(\cdot)$, and predicts $\mathbf{u}_t^T$ from $(\mathbf{z}_t^{\text{dyna}}, \mathbf{p}_t^{\text{hand}}, \mathbf{p}_t^{\text{arm}}, \mathbf{b}_t, \mathbf{f}_t, \hat{\boldsymbol{\tau}}_t)$. This asymmetric design lets the critic exploit richer simulator-only signals, while the actor refines the synthesized motion using structured kinematic and contact cues.

The reward comprises object-tracking, fingertip and wrist-guidance, and contact-encouragement terms, with penalties on unintended collisions and excessive residual signal changes (full definitions in Appendix A.1).

Deployable Student Contact and Dynamics Adaptation. Privileged variables are unavailable at deployment, so we distill a student policy from onboard observations only. We maintain two temporal history buffers over the past $K = 30$ steps: a hand-centric buffer $\mathbf{H}_t^{\text{hand}}$ stacking $\{\mathbf{p}_t^{\text{hand}}\}$ and a full-body buffer $\mathbf{H}_t^{\text{full}}$ stacking $\{\mathbf{p}_t^{\text{arm}}; \mathbf{p}_t^{\text{hand}}\}$.

A Gated Recurrent Unit (GRU) [26]-based contact adaptation module predicts $(\hat{\mathbf{b}}_t, \hat{\mathbf{f}}_t) = \phi_{\text{con}}(\mathbf{H}_t^{\text{hand}})$ and a dynamics adaptation module infers $\hat{\mathbf{z}}_t^{\text{dyna}} = \phi_{\text{dyn}}(\mathbf{H}_t^{\text{full}})$. The student actor, initialized from the teacher, predicts $\mathbf{u}_t^S$ from $(\hat{\mathbf{z}}_t^{\text{dyna}}, \mathbf{p}_t^{\text{hand}}, \mathbf{p}_t^{\text{arm}}, \hat{\mathbf{b}}_t, \hat{\mathbf{f}}_t, \hat{\boldsymbol{\tau}}_t)$ and is trained with

$$\mathcal{L}_{\text{student}} = \mathcal{L}_{\text{imitate}} + \mathcal{L}_{\text{dyna}} + \mathcal{L}_{\text{contact}}, \quad (6)$$

where $\mathcal{L}_{\text{imitate}} = \|\mathbf{u}_t^S - \mathbf{u}_t^T\|_2^2$, $\mathcal{L}_{\text{dyna}} = \|\hat{\mathbf{z}}_t^{\text{dyna}} - \mathbf{z}_t^{\text{dyna}}\|_2^2$, and $\mathcal{L}_{\text{contact}} = \lambda_b \text{BCE}(\hat{\mathbf{b}}_t, \mathbf{b}_t) + \lambda_f \|\hat{\mathbf{f}}_t - \mathbf{f}_t\|_2^2$. This enables the student to recover privileged refinement behavior from proprioceptive history alone, supporting deployment under real-world physical variations.

4 Experiments and Results

We design our experiments to answer four key questions, each corresponding to a core component of DexSynRefine: **(Q1)** Does HOI-MMFP synthesize human-object interaction (HOI) trajectories that are smoother, more spatially generalizable, and more effective as references for downstream RL compared to existing generative baselines? **(Q2)** Which action representation is most effective for grounding these HOI references into physically feasible robot behaviors? **(Q3)** Does jointly adapting object dynamics and contact states from proprioceptive history outperform single-modality adaptation? **(Q4)** Does the full DexSynRefine pipeline successfully transfer to a real robot system?

To address these questions, we evaluate our approach across the five dexterous manipulation tasks. Unless otherwise specified, all simulation results report the mean and root mean square error evaluated across 100 environments, 10 trials, and 10 random seeds. Performance metrics include the success rate (SR) at termination, as well as the object translation (OTE, in meters) and orientation (OOE, in degrees) errors averaged over the entire episode.

4.1 Motion Synthesis Quality and Downstream RL Usability

Baselines. We compare HOI-MMFP against a Task-Conditioned VAE (TC-VAE) [27], which generates trajectories by combining a task-independent latent representation with explicit task conditioning, and a full-trajectory variant (DiT-Full) that applies our flow-matching backbone directly to the trajectory space, ablating the learned motion manifold. Full architectural and training details for all three priors are provided in Appendix B. We evaluate (i) spatial generalization under initial-pose perturbations, and (ii) motion smoothness and its utility for downstream RL refinement.

Spatial Perturbation Robustness. Since our augmentation enforces a canonical initial hand configuration, a well-trained prior must start at this canonical pose regardless of object placement. We thus sweep the initial object pose along x, y (± 20 cm, 1 cm resolution) and yaw ($\pm 30^\circ$, 5° resolution), and measure the first-frame wrist error against the canonical reference. As shown in Table 1, HOI-MMFP yields the lowest error (0.015 m / 2.50°), outperforming DiT-Full (0.018 m / 2.65°) and drastically improving over TC-VAE (0.121 m / 12.88°). Figure 3(a) further shows that HOI-MMFP and DiT-Full preserve a consistent canonical start under out-of-distribution placements,

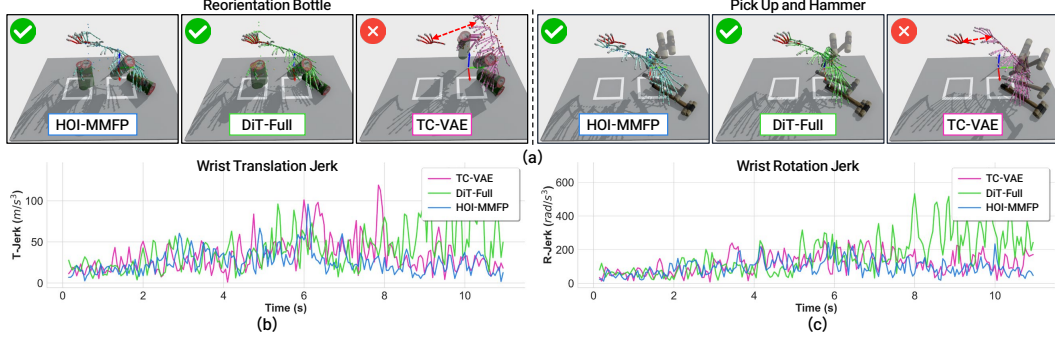


Figure 3: Qualitative comparison of HOI synthesis models. (a) Generated object and wrist trajectories visualized for two tasks. (b, c) Translational and rotational wrist jerk over time on the *Pick Up and Hammer* task. Full results are provided in Figure 7.

Method	Trajectory Prior (avg.)		Downstream Residual RL		
	Pos Err ↓	Rot Err ↓	SR (%) ↑	OTE ↓	OOE ↓
TC-VAE [27]	0.121 ± 0.068	12.88 ± 11.02	—	—	—
DiT-Full	0.018 ± 0.010	2.65 ± 1.54	49.10 ± 4.12	0.057 ± 0.006	51.42 ± 2.54
HOI-MMFP (Ours)	0.015 ± 0.010	2.50 ± 1.71	52.60 ± 3.04	0.041 ± 0.003	46.29 ± 1.39

Table 1: Synthetic trajectory quality (left, averaged over 5 tasks) and downstream residual RL performance on *Pick Up and Hammer* (right).

while TC-VAE generates distorted trajectories with shifted starting poses—likely because the latent loses semantic grounding when only the decoder is used at inference.

Motion smoothness and downstream RL utility. We assess trajectory smoothness via the mean-squared jerk of the wrist pose, computed over the trajectories in Figure 3(a) and shown over time in Figure 3(b), (c). HOI-MMFP yields the lowest translational and rotational jerk ($29.83 \pm 4.78 \text{ m/s}^3$, $101.08 \pm 17.12 \text{ rad/s}^3$), outperforming TC-VAE ($37.00 \pm 8.52 / 139.14 \pm 45.21$) and DiT-Full ($41.03 \pm 8.23 / 189.52 \pm 33.82$), which is particularly prone to jerky motion when generating in the full trajectory space without a manifold constraint. In simulation on *Pick Up and Hammer* (representative of contact-rich tasks), HOI-MMFP outperforms DiT-Full on all metrics (SR 52.6% vs. 49.1%, OTE 0.041 vs. 0.057 m, OOE 46.3° vs. 51.4°; Table 1 right). TC-VAE is excluded as its generated trajectories exhibit severe geometric distortions compared to the reference motions, preventing the RL from converging to a non-trivial success rate.

4.2 Comparing Action Representations for Leveraging the HOI Reference

Baselines. Holding the synthesized HOI prior and the reward design fixed, we compare five action representations: (i) **Joint-Space Absolute**, (ii) **Joint-Space Residual**, (iii) **Task-Space Absolute**, (iv) **Task-Space Residual (Ours)**, and (v) **Object-Centric Dexterous**, a hybrid baseline following [13] that applies task-space residuals to the wrist while predicting absolute joint commands for the fingers. We additionally include a **Kinematic Retargeting** baseline that executes the synthesized HOI reference through inverse kinematics with no learned refinement, isolating the contribution of physical grounding itself. Detailed formulations of each representation are provided in Appendix D.

Results. Table 2 reports SR, OTE, and OOE, while Figure 4 shows object-tracking reward curves. Kinematic retargeting nearly fails everywhere ($\text{SR} \leq 5.8\%$), confirming that HOI references are not directly executable. Absolute baselines generally struggle: *Task Abs* fails universally by destabilizing wrist-object alignment, and *Joint Abs* only succeeds on simple pick-and-place tasks (*Book*, *Bowl*). While *Object-Centric* is competitive, our **Task-Space Residual** achieves the highest mean SR (68.1%) and dominates every contact-rich task (*Pringles*, *Hammer*, *Watering Can*) with the most efficient learning curves. Although *Joint Abs* slightly outperforms ours on *Book* (SR 64.4% vs. 61.6%, OTE 0.035 vs. 0.054, OOE 20.0° vs. 24.7°) due to a larger direct action space in low-

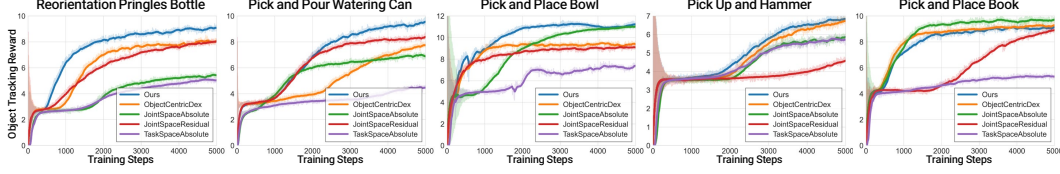


Figure 4: Object tracking reward over training steps across five tasks.

	Pringles			Watering Can			Bowl			Hammer			Book		
Action	SR	OTE	OOE	SR	OTE	OOE	SR	OTE	OOE	SR	OTE	OOE	SR	OTE	OOE
Kinematic Retar.	5.8	0.120	88.8	0.0	0.300	65.1	2.2	0.147	20.0	0.0	0.190	101.8	0.0	0.283	64.8
Joint Abs	1.4	0.119	39.1	4.1	0.130	40.9	89.0	0.032	10.8	13.0	0.151	87.9	64.4	0.035	20.0
Joint Res	51.4	0.071	43.4	43.9	0.113	27.5	7.1	0.092	24.3	1.0	0.169	84.7	41.5	0.081	30.5
Task Abs	0.0	0.134	78.5	0.0	0.132	51.0	23.1	0.143	28.4	18.0	0.131	65.1	0.5	0.144	41.9
Object-Centric [13]	66.4	0.050	33.3	19.5	0.155	50.3	57.7	0.055	21.1	52.7	0.083	55.4	60.4	0.045	23.8
Task Res (Ours)	71.5	0.044	26.5	52.4	0.107	34.2	94.8	0.023	6.8	60.3	0.053	50.0	61.6	0.054	24.7

Table 2: Comparison of action representations on five tasks.

contact regimes, its collapse on all contact-rich tasks ($\leq 13\%$ SR) demonstrates that the geometric constraints of task-space residuals are essential. Ultimately, our stable wrist anchor enables finger fine-tuning to maximize precision while inheriting the reference’s structural priors.

4.3 Effectiveness of Coupled Contact and Dynamics Adaptation

Baselines. To evaluate the individual contributions of contact and dynamics on the two most contact-rich tasks—*Pick Up and Hammer* and *Pick and Pour Watering Can*, Table 4—we compare our **Full (Ours)** method against two ablations: **w/o Contact** (adapting only dynamics, equivalent to standard RMA [21]) and **w/o Dynamics** (adapting only contact).

Results. The **Full** teacher achieves the highest success rate and lowest errors while both ablations nearly fail, showing that contact and dynamics information are jointly indispensable. During distillation, the **Full** student policy continues to yield the best overall performance. Notably, while the student nearly matches the teacher on *Watering Can* (51.8% vs. 52.4%), a gap emerges on *Hammer* (44.3% vs. 60.3%), which we attribute to the complex contact and dynamics during the pre-swing realignment phase being harder to recover from proprioceptive history alone.

4.4 Real-World Performance Evaluation

Real World Experimental Setup. We evaluate the full DexSynRefine pipeline on a real robot across all five tasks, reporting the success rate over 10 trials per task. Our platform consists of a 7-DoF arm with a 16-DoF dexterous hand running closed-loop at 20 Hz; the initial object pose required by HOI-MMFP is estimated by FoundationPose from a calibrated RGB-D camera, queried once at episode start. We compare our approach against a kinematic retargeting baseline, which directly executes the synthesized HOI-MMFP reference via inverse kinematics without any learned residuals.

Results. Table 4 reports the real-world success rates across all five tasks. Kinematic retargeting achieves at most 2/10 successful trials, failing completely (0/10) on the *Watering Can* and *Book* tasks. This confirms that directly executing synthesized HOI references is insufficient for real-world deployment due to complex physical dynamics. In contrast, integrating our residual refinement (**Residual RL**) effectively bridges this gap (as visualized in Figure 5), yielding absolute improvements of 50 to 70 percentage points across every task. Notably, our approach reaches a 9/10 success rate on Bowl and 8/10 on Pringles, demonstrating its robustness in physical environments; Notably, remaining failure modes—finger entanglement during release (*Watering Can*) and toppling during upright reorientation (*Book*)—identify contact clearance and dynamic balancing as the primary sources of the sim-to-real gap.

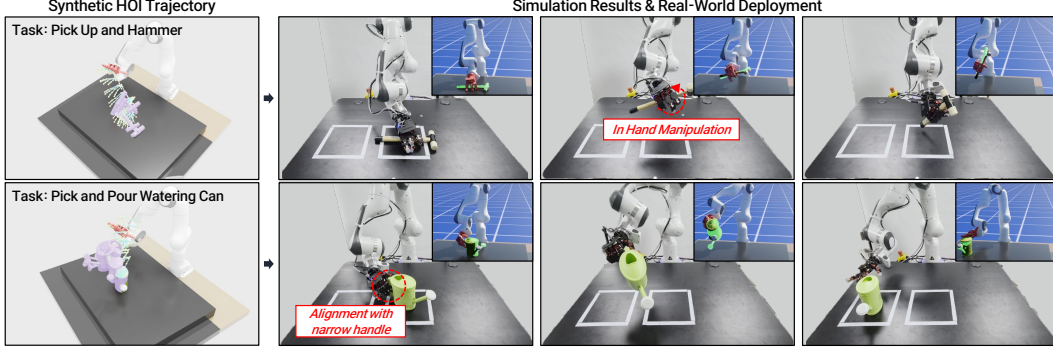


Figure 5: **Real-world and simulation deployment.** Synthetic HOI trajectories (left) and corresponding image sequences from both simulation and real-robot executions (right), shown for *Pick and Hammer* (top) and *Pick and Pour Watering Can* (bottom). See Figure 10 for remain tasks.

Stage	Adaptation	Hammer			Watering Can			Task	Kin. Retarget.	Residual RL
		SR	OTE	OOE	SR	OTE	OOE			
Teacher	Full (Ours)	60.3	0.053	50.0	52.4	0.107	34.2	Pringles	2/10	8/10
	w/o Contact	23.4	0.152	87.2	0.1	0.243	72.2	Hammer	1/10	7/10
	w/o Dynamics	9.3	0.168	97.0	5.6	0.219	67.0	Watering Can	0/10	5/10
Student	Full (Ours)	44.3	0.068	54.7	51.8	0.111	36.4	Bowl	2/10	9/10
	w/o Contact	17.2	0.163	92.5	0.0	0.228	73.4	Book	0/10	5/10
	w/o Dynamics	7.5	0.175	103.6	4.4	0.223	67.8			

Table 3: Adaptation ablation study on the *Pick and Hammer* and *Pick and Pour Watering Can* tasks.

Table 4: Real-world success rate.

5 Conclusion

We presented **DexSynRefine**, a framework that converts sparse HOI demonstrations into deployable dexterous robot behavior through three coupled components: HOI-MMFP, a task- and state-conditioned generative manifold; task-space residual RL that physically grounds synthesized references under embodiment and contact constraints; and a contact-and-dynamics adaptation module enabling sim-to-real transfer from proprioception alone. Across five tasks spanning pick-and-place, tool use, and object reorientation, our pipeline outperforms prior action representations in simulation and transfers successfully to a real robot on all five tasks, improving over kinematic retargeting by 50–70 percentage points. These results suggest that combining structured generative motion priors with task-space residual refinement is a promising path toward learning dexterous manipulation from sparse human data.

6 Limitations

We identify three key limitations of DexSynRefine. **(i) Per-task residual policies.** Because each task involves distinct contact patterns and object dynamics, we currently train a separate residual policy for each task. Future work must secure generalization capabilities to provide a single residual policy applicable to trajectories across all tasks. **(ii) Lossy contact reconstruction at deployment.** Lacking tactile sensors, the student infers contact states ($\hat{\mathbf{b}}_t, \hat{\mathbf{f}}_t$) from proprioception alone. This fails to capture rapid contact transitions, resulting in a 16 pp performance gap on the *Hammer* task (Sec 4.3). Equipping the hand with tactile fingertips would directly close this gap. **(iii) Episode-start-only conditioning of the prior.** HOI-MMFP synthesizes the trajectory $\hat{\tau}$ only once from the initial observation $\mathbf{o}_0$. Thus, if the object is displaced mid-execution—such as from external perturbations—the prior becomes stale and the system cannot reactively re-plan. Integrating HOI-MMFP with a Vision-Language Model (VLM) to perform periodic re-planning would allow the system to dynamically provide and update the target object trajectory.

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Appendix

A Details of HOI Data Collection and Trajectory Augmentation

A.1 HOI Data Collection System

Figure 6 shows our HOI data collection setup and the corresponding Open3D visualization. We attached three ArUco markers to the table to define a fixed world frame. For wrist tracking, an ArUco marker cube was attached to the wrist region of the Manus glove. Object poses were estimated from two external cameras using FoundationPose and transformed into the shared world frame.

The calibration aims to estimate the fixed rigid transformation from the Manus hand skeleton frame S to the wrist marker cube frame W , denoted as ${}^W T_S$. For this, we collected approximately 30 static samples with different hand poses. For each sample j , we logged the fingertip position p_j^S in the Manus skeleton frame, the corresponding calibration point p_j^C in the camera frame C , and the wrist cube pose ${}^C T_W(j)$. The camera-observed point was first transformed into the wrist cube frame:

$$p_j^W = ({}^C T_W(j))^{-1} p_j^C. \quad (7)$$

We then solve a robust rigid registration problem:

$$\arg \min_{R,t} \sum_{j=1}^N \rho \left(\|p_j^W - (Rp_j^S + t)\|^2 \right). \quad (8)$$

where R and t define ${}^W T_S$, and $\rho(\cdot)$ is a robust loss. The optimization is initialized using SVD-based rigid alignment. Outliers are removed based on the initial residuals, and the transformation is refined using IRLS with a Huber loss. The final calibration accuracy is validated using RMSE between the transformed Manus fingertip positions and the corresponding points in the wrist cube frame.

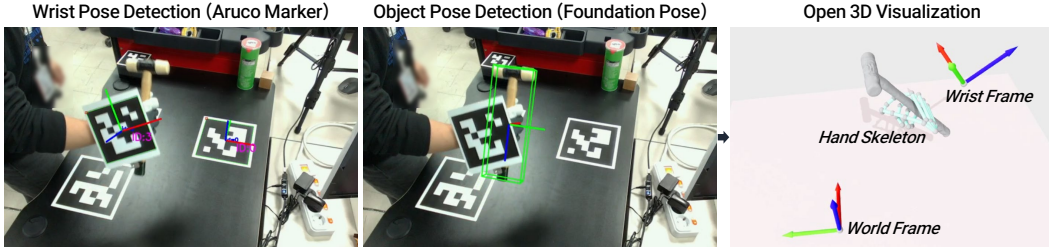


Figure 6: Overview of the HOI data collection system.

A.2 Object-Centric Trajectory Augmentation

We augment sparse HOI demonstrations in an object-centric manner to improve generalization to unseen initial object poses. We follow the notation in Sec. 3.1, where $\mathbf{w}_t$, $\mathbf{o}_t$, and $\mathbf{h}_t$ denote the wrist pose, object pose, and hand keypoints in the wrist frame, respectively. The corresponding world-frame hand keypoints are computed as

$$\mathbf{x}_{t,k} = R(\mathbf{w}_t)\mathbf{h}_{t,k} + p(\mathbf{w}_t), \quad k = 1, \dots, K. \quad (9)$$

We first remove the redundant pre-contact segment. The first interaction frame is detected by the wrist-object distance:

$$t_c = \min_t \{ \|p(\mathbf{o}_t) - p(\mathbf{w}_t)\| < \delta_c \}. \quad (10)$$

All frames before t_c are discarded.

To generate diverse initial object poses, we sample a planar object-centric transformation

$$\mathbf{G} = \begin{bmatrix} R_z(\theta) & \mathbf{b} \\ 0 & 1 \end{bmatrix}, \quad (11)$$

where $R_z(\theta)$ is a yaw rotation about the world z -axis. Given the initial object position $\mathbf{p}_0 = p(\mathbf{o}_1)$ and a sampled planar translation $\mathbf{d} = [d_x, d_y, 0]^\top$, we set

$$\mathbf{b} = \mathbf{p}_0 + \mathbf{d} - R_z(\theta)\mathbf{p}_0. \quad (12)$$

The wrist and object trajectories are transformed together:

$$\mathbf{w}'_t = \mathbf{G}\mathbf{w}_t, \quad \mathbf{o}'_t = \mathbf{G}\mathbf{o}_t. \quad (13)$$

Since the same transformation is applied to both poses, the relative hand-object geometry is preserved.

The reaching phase is regenerated from a fixed canonical hand start pose $\mathbf{w}_{\text{start}}$ to the first augmented interaction pose $\mathbf{w}'_1$:

$$\mathbf{w}_\alpha^{\text{reach}} = \text{Interp}(\mathbf{w}_{\text{start}}, \mathbf{w}'_1; \alpha), \quad \alpha \in [0, 1]. \quad (14)$$

Translation is linearly interpolated, while rotation is interpolated using slerp. The hand keypoints in the wrist frame are also linearly interpolated from the canonical initial hand configuration to $\mathbf{h}_1$.

For the object-use phase, we preserve the original object-to-wrist relation:

$$\mathbf{T}_t^{ow} = \mathbf{o}_t^{-1}\mathbf{w}_t. \quad (15)$$

Given the augmented object trajectory $\hat{\mathbf{o}}_t$, the wrist trajectory is reconstructed as

$$\hat{\mathbf{w}}_t = \hat{\mathbf{o}}_t \mathbf{T}_t^{ow}. \quad (16)$$

This retargets the hand motion to the augmented object pose while maintaining the original grasp relation.

For axis-symmetric objects, we additionally apply a random local-axis rotation to the object orientation:

$$R(\hat{\mathbf{o}}_t) \leftarrow R(\hat{\mathbf{o}}_t)R_a(\phi), \quad \phi \sim \mathcal{U}(-\pi, \pi), \quad (17)$$

where $R_a(\phi)$ is a rotation around the object's local symmetry axis. Finally, all augmented trajectories are resampled to a fixed length using linear interpolation for translations and hand keypoints, and slerp for rotations.

B Details of Generative Motion Prior Baselines

This section describes the implementation of HOI-MMFP and the two generative motion-prior baselines (TC-VAE and DiT-Full) used in Sec. 4.1. To isolate the effect of *where* generation happens (latent vs. trajectory space) and *how* the prior is trained (one-stage vs. two-stage), all three methods share the same trajectory representation, conditioning interface, and training schedule; method-specific differences are restricted to the generative formulation.

B.1 Shared Setup

Trajectory representation. Each episode is a fixed-length sequence $\tau \in \mathbb{R}^{T \times D}$ with $T=220$ and per-frame features $D=93$: $[\mathbf{w}_t^{\text{pos}}(3), \mathbf{w}_t^{\text{rot6d}}(6), \mathbf{o}_t^{\text{pos}}(3), \mathbf{o}_t^{\text{rot6d}}(6), \mathbf{h}_t^{\text{local}}(75)]$. Rotations use the 6D representation; trajectories are normalized per-feature using training-set statistics.

Conditioning. All methods receive the same task condition $\mathbf{c} = (\mathbf{u}, \mathbf{o}_0)$. The numeric component $\mathbf{o}_0 \in \mathbb{R}^9$ is the initial object pose (3D position + 6D rotation), and the semantic component $\mathbf{u} \in \mathbb{R}^{384}$ is obtained from the frozen all-MiniLM-L6-v2 encoder [28] (mean-pooled, ℓ_2 -normalized). Inside each network, $\mathbf{o}_0$ and $\mathbf{u}$ are projected by separate two-layer MLPs and added to the conditioning vector.

Training. We use identical optimization for all methods: fused AdamW (lr 3×10^{-4} , wd 10^{-4} , grad clip 1.0), 1,000 epochs, batch size 256, bf16 mixed precision, and the same random seed and batch ordering. The augmented HOI dataset (canonical wrist start, randomized object x, y , yaw) is split 95/5 into train/val. The shared backbone uses hidden dim $d_h=256$ with 3 Transformer layers and 4 attention heads.

B.2 HOI-MMFP

Stage 1: Motion manifold AE. A 3-layer Transformer encoder ($d_h=256$, 4 heads, GELU, dropout 0.1) maps τ^{aug} to $\mathbf{z}^* \in \mathbb{R}^{16}$ via mean pooling and a LayerNorm–Linear head. The decoder broadcasts $\mathbf{z}^*$ over $T=220$ time tokens (after a 2-layer MLP expansion), adds positional embeddings, and applies a 3-layer Transformer followed by a linear head. Loss weights: $\eta=10^{-3}$, $\delta=10^{-2}$, $\alpha \sim \mathcal{U}(-0.4, 1.4)$.

Stage 2: Latent conditional flow. With the AE frozen, a velocity field v_θ operates on a sequence of 16 scalar tokens (one per latent coordinate), each embedded to $d_h=256$. Three adaLN-Zero DiT blocks [24] (4 heads, MLP ratio 4) are modulated by the sum of (i) a sinusoidal time embedding (dim 256, scale 10^3), (ii) projected $\mathbf{o}_0$, and (iii) projected $\mathbf{u}$. Inference uses a 50-step Heun ODE solver to integrate $d\mathbf{z}_s/ds=v_\theta(\mathbf{z}_s, s, \mathbf{c})$ from $s=0$ to $s=1$, after which $\hat{\tau}=D(\hat{\mathbf{z}})$.

B.3 TC-VAE Baseline

We follow the original Task-Conditioned VAE formulation [27]: the latent $\mathbf{h} \in \mathbb{R}^{16}$ encodes *manner* variation that must be statistically independent of the specified *task* parameters $\mathbf{w} = \mathbf{c}$. We replace the original TCN backbone with a Transformer encoder–decoder that matches HOI-MMFP in capacity, and keep the rest of the recipe unchanged. The encoder $q_\phi(\mathbf{h}|\tau)$ outputs a diagonal Gaussian, and the decoder $p_\theta(\tau|\mathbf{h}, \mathbf{c})$ is conditioned on both $\mathbf{h}$ and $\mathbf{c}$. Training combines a β -VAE ELBO with an adversarial information objective: an auxiliary 3-layer MLP f_W predicts $\mathbf{c}$ from $\mu_\phi(\tau)$, and the encoder is trained to fool it,

$$\min_{\theta, \phi} \mathcal{L}_{\text{ELBO}} - \alpha \mathcal{L}_{\text{aux}}, \quad \min_W \mathcal{L}_{\text{aux}} = \|f_W(\mu_\phi(\tau)) - \mathbf{c}\|_1, \quad (18)$$

where $\mathcal{L}_{\text{ELBO}} = \beta \text{KL}(q_\phi\|\mathcal{N}(0, I)) - \mathbb{E}_q[\log p_\theta]$. Following [27] we set $\beta=8$ and tune α on the validation set. At inference we draw $\mathbf{h} \sim \mathcal{N}(0, I)$ and decode under $\mathbf{c}$. Because reconstruction does not enforce any condition-equivariant structure on the latent, the decoder must re-derive spatial grounding from $\mathbf{c}$ alone, which is brittle under out-of-distribution $\mathbf{o}_0$ (cf. Table 1).

B.4 DiT-Full Baseline

DiT-Full ablates the manifold by applying our flow-matching backbone directly to the raw trajectory: tokens are time steps rather than latent coordinates, and the input $\mathbf{x}_s \in \mathbb{R}^{T \times D}$ is linearly projected to $d_h=256$ with $T=220$ learned positional embeddings. Three DiT blocks (4 heads, MLP ratio 4) modulated by the same $(s, \mathbf{o}_0, \mathbf{u})$ produce a per-frame velocity $v_\theta \in \mathbb{R}^{T \times D}$. Training uses the rectified flow-matching loss with $\mathbf{x}_s=(1-s)\mathbf{x}_0 + s\tau^{\text{aug}}$, and inference uses the same 50-step Heun integrator. DiT-Full and HOI-MMFP therefore differ *only* in the space the flow operates on, providing a controlled ablation of the learned motion manifold.

B.5 Qualitative Results and Latent Structure

Generated trajectories across all tasks. Figure 7 visualizes HOI-MMFP outputs on all five tasks. The generated wrist–object–finger trajectories are temporally smooth, remain faithful to the canonical initial pose enforced during augmentation, and adapt their shape to each task’s text condition.

Latent organization under spatial perturbation. To inspect why the latent flow generalizes across the perturbation grid of Sec. 4.1, we sample $\hat{\mathbf{z}}$ under each perturbed initial object pose (x, y) swept at 1 cm resolution within ± 20 cm; yaw at 5° within $\pm 30^\circ$ and project the latents to 2D with UMAP [29]. Figure 8 reveals two levels of structure. At the coarse level, the latents form distinct clusters, one per task, showing that the conditional flow first separates samples by task identity. Within each task cluster, perturbations along x , y , and yaw further trace axis-aligned variations, indicating that spatial perturbations are arranged meaningfully along the corresponding directions in latent space rather than collapsing into noise. The coexistence of global task-level clustering and

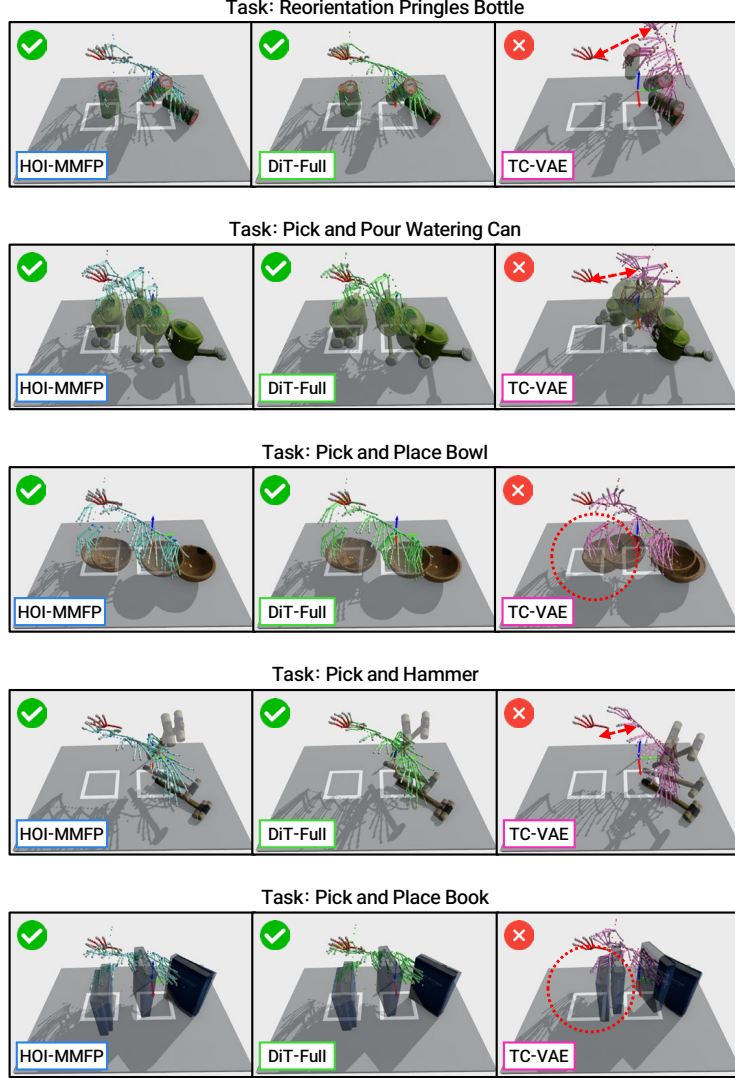


Figure 7: Generated HOI trajectories for all tasks. The red hand denotes the canonical hand pose.

locally axis-aligned spatial structure suggests that flow matching, when carried out on the learned manifold, organizes the conditional generative process in a geometrically faithful way.

C Implementation Details of Residual Action RL Policy Training

C.1 Teacher Reward Design

The teacher reward is designed to preserve the kinematic structure of the synthesized HOI reference while ensuring physical feasibility under the robot’s embodiment. The total reward r_t is a weighted sum of five components,

$$r_t = \omega_{\text{obj}} r_{\text{obj}} + \omega_{\text{hand}} r_{\text{hand}} + \omega_{\text{wrist}} r_{\text{wrist}} + \omega_{\text{contact}} r_{\text{contact}} + \omega_{\text{penalty}} r_{\text{penalty}}, \quad (19)$$

where $d_p(\cdot, \cdot)$ and $d_r(\cdot, \cdot)$ denote the Euclidean position distance and angular rotation distance, and $a_p^{(\cdot)}, a_r^{(\cdot)}$ are positive scaling coefficients. The fingertip and wrist tracking terms, $r_{\text{hand}} = \exp(-a_p^h d_p(\mathbf{h}_t, \hat{\mathbf{h}}_t))$ and $r_{\text{wrist}} = \exp(-a_p^w d_p(\mathbf{w}_t, \hat{\mathbf{w}}_t)) + \exp(-a_r^w d_r(\mathbf{w}_t, \hat{\mathbf{w}}_t))$, are active throughout the entire episode and supervise the kinematic refinement of the synthesized reference. The object tracking reward, in contrast, is conditionally activated so that the policy is not penalized

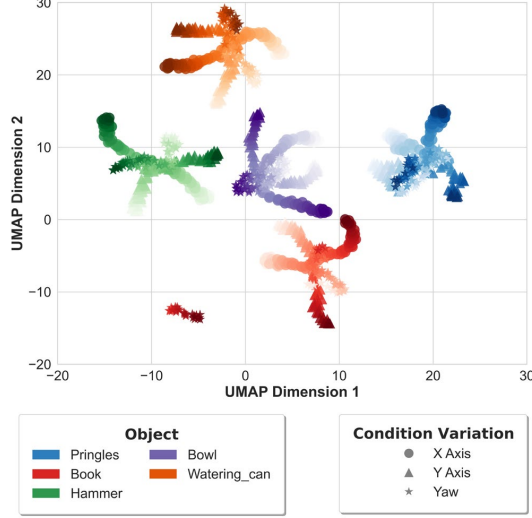


Figure 8: Conditional flow matching latent UMAP projection.

for failing to track the object pose during the reaching phase, when the hand has not yet established a viable grasp configuration. Concretely, letting $\mathbf{p}_t^{\text{palm}}$ denote the palm position, $\mathbf{c}(\mathbf{o}_t)$ the object center, and $R(\mathbf{o}_t)$ the radius of the smallest object-aligned bounding sphere, we activate the term once the proximity condition $\|\mathbf{p}_t^{\text{palm}} - \mathbf{c}(\mathbf{o}_t)\|_2 \leq R(\mathbf{o}_t) + \delta$ with $\delta = 10$ cm is first satisfied at t^* , and define

$$r_{\text{obj}} = \begin{cases} \exp(-a_p^o d_p(\mathbf{o}_t, \hat{\mathbf{o}}_t)) + \exp(-a_r^o d_r(\mathbf{o}_t, \hat{\mathbf{o}}_t)), & t \geq t^*, \\ 0, & t < t^*. \end{cases} \quad (20)$$

Once activated, the reward remains active for the rest of the episode regardless of subsequent palm-object distance, so that momentary regrasps or in-hand rolling do not deactivate object supervision. The remaining terms regulate physical interaction. The contact reward $r_{\text{contact}} = \|\mathbf{b}_t\|_1$ counts the number of active binary contact sensors on the hand, encouraging stable multi-finger contact during manipulation. The penalty term $r_{\text{penalty}} = p_{\text{col}} + p_{\text{reg}}$ aggregates a collision penalty $p_{\text{col}} = -\sum_{i \in \mathcal{S}_{\text{fin}}} \|\mathbf{f}_{t,i}\|_2 - \sum_{j \in \mathcal{S}_{\text{bod}}} \|\mathbf{f}_{t,j}\|_2$, where $\mathcal{S}_{\text{fin}}$ and $\mathcal{S}_{\text{bod}}$ are the sets of contact sensors on non-fingertip hand links and arm body segments respectively, and a smoothness regularization $p_{\text{reg}} = -a_v \|\dot{\mathbf{q}}_t\|_2$ that suppresses excessive joint velocities. The collision penalty protects the hardware and prevents the policy from exploiting spurious collisions during training, while the regularization encourages smooth, deployable motion.

C.2 Policy Training Details

The teacher policy, which learns residual actions for refining the synthesized HOI motion, is trained in IsaacLab [30] using PPO. We use 8192 parallel environments for teacher policy training and 4096 parallel environments for student policy distillation. The reward weights and scaling coefficients used for teacher policy learning are summarized in the left panel of Table 5, while the PPO hyperparameters are reported in the right panel of Table 5. The task-space residual action integration parameters are provided in Table 6. Detailed observation notation and dimensions are listed in Table 8, and the network architecture and adaptation module configuration are summarized in Table 7. The domain randomization settings used during simulation training are reported in Table 9.

D Details of Action Representation Baselines

We summarize the action representations used as baselines in Sec. 4.2. For all baselines, the synthesized HOI trajectory provides a reference wrist pose and hand target, denoted by $\hat{\tau}_t$. The policy action is normalized to $[-1, 1]$ and converted into either joint-space or task-space commands. All

Symbol	Value	PPO Parameter	Value
ω_{obj}	20.0	Value loss coef.	1.0
ω_{hand}	5.0	Clipped value loss	True
ω_{wrist}	2.0	Clip parameter	0.2
ω_{contact}	1.0	Entropy coef.	0.005
ω_{penalty}	0.1	Learning epochs	5
a_p^o, a_r^o	40.0, 0.5	Mini-batches	8
a_p^h	10.0	Learning rate	5.0×10^{-4}
a_p^w, a_r^w	20.0, 2.0	Schedule	Adaptive
a_v	0.3	γ	0.99
δ	0.10 m	λ	0.95
		Desired KL	0.01
		Max grad. norm	1.0

Table 5: Reward coefficients and PPO hyperparameters used for teacher policy training.

Parameter	Symbol	Value
EMA coefficient	α	0.8
Policy timestep	Δt	0.05 s
Wrist position scale	s_p^w	0.02 m
Wrist rotation scale	s_r^w	5°
Fingertip position scale	s_p^f	0.03 m
Wrist position residual limit	$\Delta \mathbf{a}_{\min/\max}^{w,p}$	± 0.10 m
Wrist rotation residual limit	$\Delta \mathbf{a}_{\min/\max}^{w,r}$	$\pm 40^\circ$
Fingertip position residual limit	$\Delta \mathbf{a}_{\min/\max}^{f,p}$	± 0.05 m

Table 6: Task-space residual integration parameters.

resulting targets are clipped to the corresponding joint or task-space limits before being sent to the inverse kinematics solver or joint controller.

Object-Centric Baseline Action. The object-centric baseline uses a mixed action representation. The policy predicts a wrist residual command and an absolute hand joint command:

$$\mathbf{u}_t = [\mathbf{u}_t^w, \mathbf{u}_t^h], \quad \mathbf{u}_t^w \in \mathbb{R}^6, \quad \mathbf{u}_t^h \in \mathbb{R}^{16}. \quad (21)$$

The wrist residual is integrated in task space around the synthesized wrist reference:

$$\Delta \mathbf{a}_t^w = \text{clip}(\Delta \mathbf{a}_{t-1}^w + s_w \tilde{\mathbf{u}}_t^w \Delta t, \Delta \mathbf{a}_{\min}^w, \Delta \mathbf{a}_{\max}^w), \quad (22)$$

where $\tilde{\mathbf{u}}_t^w$ is the EMA-filtered wrist command. The refined wrist target is obtained by applying this residual to the reference wrist pose in $\hat{\tau}_t$. The hand command is directly mapped to an absolute joint target:

$$\mathbf{q}_t^h = \text{scale}(\mathbf{u}_t^h, \mathbf{q}_{\min}^h, \mathbf{q}_{\max}^h). \quad (23)$$

The arm joint target is then obtained by solving IK from the refined wrist target.

Joint-Space Absolute Action. The joint-space absolute baseline predicts arm and hand joint commands:

$$\mathbf{u}_t = [\mathbf{u}_t^a, \mathbf{u}_t^h], \quad \mathbf{u}_t^a \in \mathbb{R}^7, \quad \mathbf{u}_t^h \in \mathbb{R}^{16}. \quad (24)$$

The arm action is applied as a velocity-like update to the previous arm joint target:

$$\mathbf{q}_t^a = \text{clip}(\mathbf{q}_{t-1}^a + s_a \mathbf{u}_t^a \Delta t, \mathbf{q}_{\min}^a, \mathbf{q}_{\max}^a). \quad (25)$$

The hand action is mapped to an absolute hand joint target:

$$\mathbf{q}_t^h = \text{scale}(\mathbf{u}_t^h, \mathbf{q}_{\min}^h, \mathbf{q}_{\max}^h). \quad (26)$$

Parameter	Value
Initial noise std.	1.0
Intrinsic encoder hidden dims.	[128, 64]
Intrinsic encoder output dim.	16
Intrinsic encoder activation	ELU
Actor hidden dims.	[1024, 512, 256, 128]
Critic hidden dims.	[1024, 512, 256, 128]
Actor-critic activation	ELU
Contact adaptation architecture	GRU-based temporal encoder
Dynamics adaptation architecture	GRU-based temporal encoder
GRU hidden dim.	32
GRU layers	2
Dynamics latent dim.	16
Binary contact threshold	0.5

Table 7: Network architecture.

Notation	Space	Description
$\mathbf{p}_t^{\text{arm}}$	$\mathbb{R}^{23}$	Arm proprioception, including arm joint positions (7), wrist position (3), wrist rotation in 6D representation (6), and previous arm action (7).
$\mathbf{p}_t^{\text{hand}}$	$\mathbb{R}^{44}$	Hand proprioception, including hand joint positions (16), fingertip positions in the world frame (12), and previous hand action (16).
$\mathbf{H}_t^{\text{hand}}$	$\mathbb{R}^{30 \times 44}$	Hand-centric proprioceptive history over the past 30 policy steps, used by the contact adaptation module.
$\mathbf{H}_t^{\text{full}}$	$\mathbb{R}^{30 \times 67}$	Full-body proprioceptive history over the past 30 policy steps, stacking both arm and hand proprioception.
$\mathbf{b}_t$	$\{0, 1\}^{11}$	Binary contact state, including distal contacts (4), medial contacts (4), and proximal contacts (3).
$\mathbf{f}_t$	$\mathbb{R}^{33}$	Normalized contact force, including distal contact forces (12), medial contact forces (12), and proximal contact forces (9).
$\mathbf{z}_t^{\text{dyna}}$	$\mathbb{R}^{16}$	Latent dynamics representation compressed by the dynamics encoder from object intrinsic parameters.
$\boldsymbol{\tau}_t$	$\mathbb{R}^{46}$	Synthesized HOI reference, including target wrist pose (3 + 6), target object pose (3 + 6), target fingertip positions (12), wrist tracking error (3 + 1), and fingertip tracking error (12).

Table 8: Notation and dimensions for deployable student contact and dynamics adaptation.

Joint-Space Residual Action. The joint-space residual baseline predicts residuals around the nominal HOI joint targets. The nominal targets are obtained from the synthesized HOI reference by IK:

$$\hat{\mathbf{q}}_t^a, \hat{\mathbf{q}}_t^h = \text{IK}(\hat{\boldsymbol{\tau}}_t). \quad (27)$$

The policy predicts residual commands for the arm and hand:

$$\mathbf{u}_t = [\mathbf{u}_t^a, \mathbf{u}_t^h], \quad \mathbf{u}_t^a \in \mathbb{R}^7, \quad \mathbf{u}_t^h \in \mathbb{R}^{16}. \quad (28)$$

After EMA filtering, the residuals are integrated and clipped:

$$\Delta \mathbf{q}_t = \text{clip}(\Delta \mathbf{q}_{t-1} + s_q \tilde{\mathbf{u}}_t \Delta t, \Delta \mathbf{q}_{\min}, \Delta \mathbf{q}_{\max}). \quad (29)$$

The final joint target is obtained by adding the residual to the nominal HOI target:

$$\mathbf{q}_t = \text{clip}(\hat{\mathbf{q}}_t + \Delta \mathbf{q}_t, \mathbf{q}_{\min}, \mathbf{q}_{\max}). \quad (30)$$

This representation constrains the policy to refine the synthesized HOI motion in joint space rather than generating the entire motion from scratch.

Parameter	Type	Distribution	Range
Robot			
Mass	Scaling	Uniform	[0.5, 1.5]
Static friction	Absolute	Uniform	[0.2, 1.0]
Dynamic friction	Absolute	Uniform	[0.2, 1.0]
Restitution	Absolute	Uniform	[0.0, 0.4]
Joint stiffness	Scaling	Uniform	[0.5, 1.5]
Joint damping	Scaling	Uniform	[0.3, 3.0]
Joint limits	Scaling	Uniform	[0.95, 1.05]
Joint friction	Scaling	Uniform	[0.8, 1.2]
Armature	Scaling	Uniform	[0.8, 1.2]
Object			
Mass	Absolute	Uniform	[0.1, 1.0] kg
Static friction	Absolute	Uniform	[0.2, 0.6]
Dynamic friction	Absolute	Uniform	[0.2, 0.6]
Restitution	Absolute	Uniform	[0.0, 0.4]
COM offset (x, y, z)	Additive	Uniform	[-0.02, 0.02] m
Uniform scale	Scaling	Uniform	[0.95, 1.05]
Desk / Partition			
Static friction	Absolute	Uniform	[0.5, 1.1]
Dynamic friction	Absolute	Uniform	[0.5, 1.1]
Restitution	Absolute	Uniform	[0.0, 0.4]
Environment			
Gravity perturbation (z)	Additive	Uniform	[0.0, 0.5] m/s ²

Table 9: Domain randomization settings used during simulation training.

Task-Space Absolute Action. The task-space absolute baseline predicts task-space velocity commands for the wrist and fingertips:

$$\mathbf{u}_t = [\mathbf{u}_t^w, \mathbf{u}_t^x], \quad \mathbf{u}_t^w \in \mathbb{R}^6, \quad \mathbf{u}_t^x \in \mathbb{R}^{4 \times 3}. \quad (31)$$

The wrist command is updated by integrating the predicted translational and rotational velocities:

$$\mathbf{a}_t^w = \mathbf{a}_{t-1}^w + s_w \mathbf{u}_t^w \Delta t. \quad (32)$$

The fingertip targets in the wrist frame are similarly updated as

$$\mathbf{x}_t^W = \mathbf{x}_{t-1}^W + s_x \mathbf{u}_t^x \Delta t. \quad (33)$$

The commanded wrist pose and fingertip positions are then converted to executable arm and hand joint commands through IK:

$$\mathbf{q}_t = \text{IK}(\mathbf{a}_t^w, \mathbf{x}_t^W). \quad (34)$$

Unlike the residual formulation, this baseline integrates commands from the previous task-space target rather than refining the synthesized HOI reference at each step.

E Implementation Details of Real-World Experiments

Hardware and workspace. Our real-world platform consists of a Franka Panda 7-DoF arm equipped with a 16-DoF dexterous hand, and RGB-D observations are captured by an Intel RealSense L515 camera (Figure 9). To evaluate the prior under realistic conditions, we use *actual real-world objects* rather than 3D-printed replicas. Two white rectangles are marked on the table: the left rectangle indicates the target placement region for pick-and-place tasks, and the right rectangle defines the initial grasping region. The grasping region is sized to coincide with the in-distribution range of object placements seen during training and augmentation.



Figure 9: Real-world experimental setup.

Object pose estimation. To obtain the object’s 6-DoF pose at every step we use FoundationPose, which requires an initial object mask. At each episode start we generate this mask by combining an open-vocabulary detection from Grounding DINO [31] with segmentation from SAM2 [32]; the resulting mask is supplied once to FoundationPose, which then tracks the object through the rest of the episode.

Policy execution. In simulation the policy runs at the simulator step rate of 120 Hz, producing task-space targets that are then quintic-interpolated. On the real robot, policy queries are issued at 20 Hz; their task-space targets are quintic-interpolated and streamed to `libfranka` at its 1 kHz real-time control loop. Because the generated reference trajectory does not always start at the robot’s current configuration, we move the robot to the initial pose of the generated trajectory before each episode and roll out the policy from there.

Additional qualitative results. Figure 10 shows real-world rollouts for the remaining tasks not included in Figure 5.

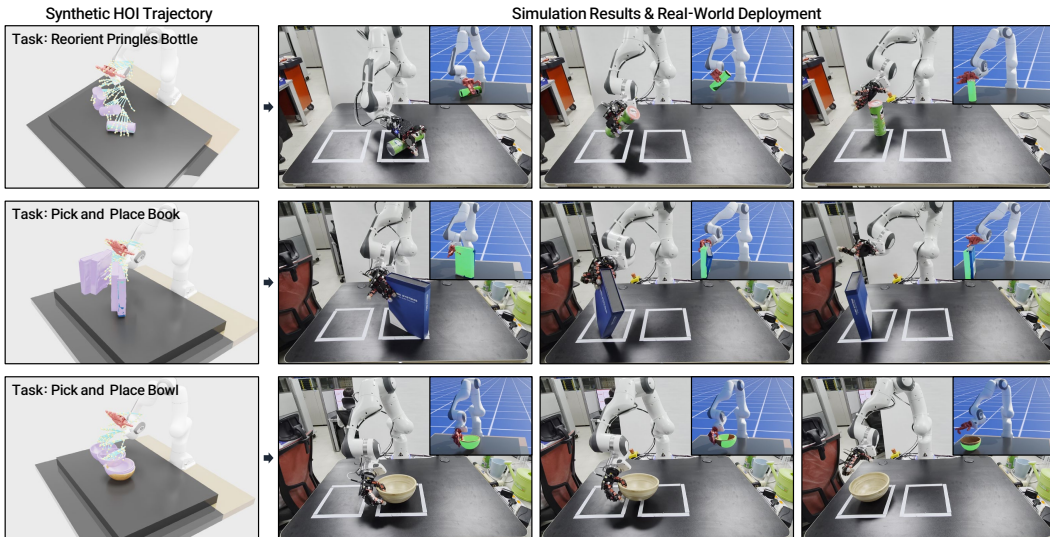


Figure 10: Real-world rollouts for the remaining tasks.

F Joint-Limit Compliance via Residual RL

IK alone violates joint limits. We deliberately choose a Franka Panda base configuration that emulates the kinematic profile of a human arm so that wrist trajectories from the human-trained prior fall within the robot’s reachable workspace. Even with this alignment, tracking the wrist reference with damped least-squares (DLS) IK frequently drives the solved joint configuration into one or more joint limits before the wrist reaches its target, causing the planner to abort. This occurs even when the wrist target itself is geometrically reachable, because DLS IK has no awareness of joint-limit headroom and will spend its redundancy arbitrarily.

Residual RL restores feasibility. Adding the residual policy on top of the same wrist reference yields task-space targets whose IK solutions remain inside the joint-limit box throughout the rollout while still completing the task. The residual therefore acts as a learned correction that absorbs the mismatch between the human-derived reference and the robot’s joint-space constraints. Figure 11 illustrates this comparison: the top row shows a simulation snapshot of the pure-**IK** rollout (left) and the corresponding per-joint trajectories over time (right), in which one or more joints saturate at their limits; the bottom row shows the residual-**RL** rollout, whose joint trajectories stay well clear of the limits while the task is executed successfully.

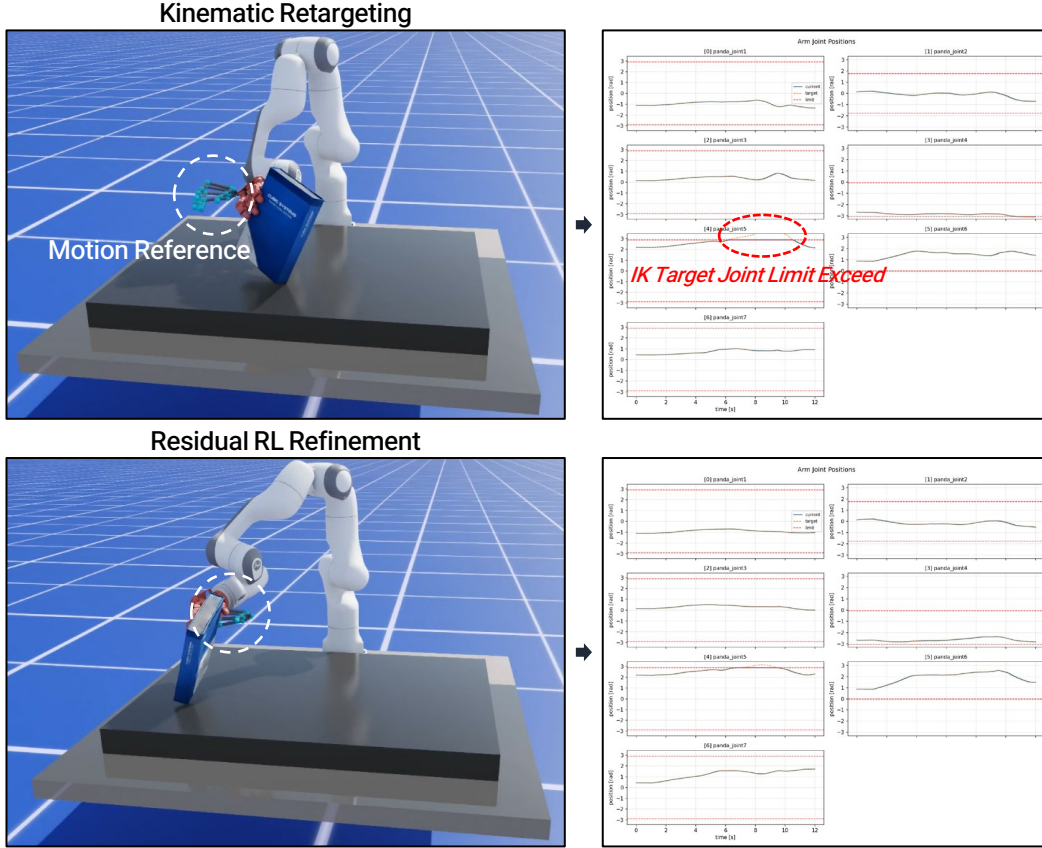


Figure 11: Joint-limit behavior under pure DLS-**IK** tracking (*top*) versus residual RL (*bottom*).